

Smart Traffic Management System for clearance of emergency vehicles using IOT

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Abstract : India's population is increasing rapidly, which has led to a tremendous increase in the number of vehicles on the road. This has made it difficult to manage emergency vehicles such as ambulances, fire brigades, and police vehicles, as there is often congestion on the roads. To address this issue, a new model has been proposed that focuses on managing traffic flow and clearing the way for emergency vehicles. Previous research has identified the priority of vehicles, but more work is needed to fully solve the problem. A proposed solution to tackle traffic congestion for emergency vehicles involves the use of IoT technology. This model involves the use of devices like HC-SR04 Ultrasonic Sensor, RC522 RFID Module, RFID tag, Ohms Resistor, Signal Light Module, and Connection Wires to manage traffic signals. When an emergency vehicle approaches, the sensors in the vehicle and signal communicate with each other to change the signal, allowing the emergency vehicle to pass through smoothly.

Keywords: Ultrasonic Sensors, RFID Technology, Signal Light modules, AurdinoIDE, Aurdino.

I. INTRODUCTION:

The issue of traffic congestion is a significant problem in developing countries like India. The objective of this project is to create an intelligent traffic control system that enables emergency vehicles, such as ambulances, fire trucks, and VIP vehicles, to reach their destination promptly. The system will provide clearance for emergency vehicles, even in congested traffic areas, ensuring their shift movement. For summarization many older systems are built using Multi Modal Biometrics. In case of proctoring, many recent developments are made using Convolutional Neural Networks. Various methods of proctoring have their own strengths and weaknesses, and they employ different techniques for monitoring.

The exiting traffic system is generally controlled by the traffic police. The traffic control system operated by traffic police has a major limitation in that it lacks intelligence to effectively manage traffic congestion. The decision-making process of the system is solely dependent on the judgement of the traffic police officials, who may choose to block a road for an extended period or allow vehicles to pass through another route. Additionally, even with the use of traffic lights, the fixed timing of green and red signals may not be sufficient in addressing the issue of traffic congestion. In India, despite the presence of traffic lights, there is still a need for human intervention, making it an uneconomical solution requiring more personnel. A new model has been suggested to alleviate traffic congestion and prioritize emergency vehicles. While some research has already been done in traffic management, it has not been enough to fully solve the issue. The proposed

solution involves utilizing the Internet of Things (IoT) technology to manage traffic signals through devices such as ultrasonic sensors, RFID modules, resistors, signal lights, and connection wires. By communicating with the sensors on the emergency vehicle, the traffic signal would change accordingly to clear the way for the vehicle. Impact, Significance, and Contribution: With Smart Traffic Management System (STMS) advancements, an ambulance will be able to obtain an open lane and arrive at the hospital on time. The proposed methodology aims to offer solutions for traffic congestion, emergency vehicle clearance, and vehicle tracking. An important benefit of this system is its versatility, as it can operate in both automatic and manual modes. When in automatic mode, the system utilizes sensor data to make decisions, whereas in manual mode, the driver assumes control.

LITERATURE REVIEW

Presenting a distinct architecture for an intelligent traffic management system is the author's main goal. The system uses sensors to track traffic density, especially at intersections, and gather information that traffic lights utilize to regulate different traffic densities. Unfortunately, this method has drawbacks, such being too expensive and unable to detect traffic density beyond a specific range. Adaptive traffic control also makes use of methods similar to these. A traffic signal control method developed to solve the management of emergency vehicle traffic utilizing IOT. In order to provide proper traffic control, this system enables

emergency vehicles to notify the traffic signal controller stationed at intersections with several lanes of their presence. By placing a sensor in each one, a traffic light control system has been created to give emergency vehicles priority. As the emergency vehicle arrives, the sensor at each intersection. emergency vehicle able to take a different route than other cars to get to its destination thanks to the receiver's demodulation of the signal code and activation of the red traffic light which is present at the intersection where the signal is present.

A survey of literature on smart traffic management systems reveals that numerous studies have explored the advantages and drawbacks of such systems. Scholars have investigated the impact of smart traffic management systems on traffic flow, safety, and environmental sustainability. Additionally, researchers have examined the technical components of these systems, such as the use of sensors, cameras, and artificial intelligence algorithms for traffic monitoring and control. Moreover, studies have analyzed the integration of smart traffic management systems with other intelligent transportation systems, such as connected and self-driving vehicles. The literature also acknowledges the challenges and limitations of smart traffic management systems, including the necessity for efficient data management, privacy and security concerns, and the high expenses of installation and maintenance. Overall, the literature survey highlights the potential of smart traffic management systems to improve transportation efficiency, safety, and sustainability, while also emphasizing the need for further research and development to overcome the remaining obstacles.

A traffic management system is in place in Cambridge city, which uses sensors buried in the roads to detect traffic queues and send information to a central control unit for decision-making. However, the system may experience delays due to networking issues since it is centralized. Researchers have experimented with using surveillance cameras and number plate recognition to identify vehicles, but this method may not be effective in countries like Pakistan where there are various types of traffic without number plates. Another system proposed by Osman et al. also utilized surveillance cameras to determine traffic density, with images being sent to a server for analysis. This system utilized fixed thresholds based on the number of vehicles on the road, and an algorithm was used to set the duration of red lights at intersections based on traffic density.

The traffic police often have responsibility over the exiting traffic system. This system, which is managed by the traffic police, has a major flaw in that it is not intelligent enough to handle traffic congestion. The traffic police officer has two options: either extend the duration that a road is blocked or let traffic on another road to pass. In other words, the official's choice may not be the wisest one. Additionally, even if traffic lights are employed, a defined time period is established for when the vehicles will display a green or red signal.

Because of this, it might not address the issue of traffic congestion. It has been observed in India that traffic police officers remain on duty even in the presence of traffic lights, indicating that this system requires more work force and is not cost-effective.

The Internet of Things (IoT), also known as the Internet of Everything (IoE), is a network of web-connected devices that collect, send, and act on data from their surroundings using embedded sensors, processors, and communication hardware. These "connected" or "smart" devices can communicate with one another via machine-to-machine (M2M) communication, allowing them to act on the data they receive. While humans can interact with these devices to configure them or access their data, they generally operate on their own. The widespread availability of mobile components and always-on networks has allowed these devices to proliferate. While they generate large amounts of data that can be useful, they also raise privacy and security concerns. However, IoT technology enables previously unavailable real-time data. For example, homes and businesses can be remotely monitored for safety and productivity can be improved, while sensors in city infrastructure can help reduce congestion and warn of potential dangers.

RELATED RESEARCH

The author describes an intelligent system for traffic management that uses sensors to monitor traffic density, especially at junctions. The data collected from these sensors, along with RFID monitoring of traffic density, is used to manage traffic lights. However, this method has limitations in identifying traffic density beyond a certain distance and is expensive. An alternative approach is adaptive traffic control. The author also proposes a traffic signal control technique using the internet of things (IOT) to improve traffic management for emergency vehicles. The suggested solution enables the emergency vehicles to communicate their arrival to the traffic signal controller stationed at the intersection of two or more lanes so that the traffic can be controlled appropriately. A traffic light control system specifically made for emergency vehicles uses a sensor in each one of them to send a signal to a receiver at each traffic light intersection. Signal code will be given to the receiver after the emergency vehicle has arrived at the intersection with the traffic lights. The receiver demodulates the code, activating the red traffic light at the intersection. Emergency vehicles will therefore travel a different path to their destination than ordin.

IMPACT SIGNIFICANCE & CONTRIBUTION

Smart traffic management systems can bring about substantial transformations to transportation networks and society at large. The ensuing are some essential benefits and contributions of smart traffic management systems:

Smart traffic management systems can enhance traffic

flow and decrease congestion by monitoring traffic patterns in real-time and dynamically adjusting traffic signals. This results in reduced time spent idling in traffic and improved travel times.

Smart traffic management systems can improve safety on roads by detecting and promptly responding to incidents in real-time, such as accidents, road closures, or adverse weather conditions. This can assist in decreasing the frequency and severity of accidents on the roads.

Smart traffic management systems can contribute to environmental sustainability by reducing congestion and optimizing traffic flow. This can lead to a decrease in carbon emissions and an improvement in air quality.

III. METHODOLOGY:

When there is no emergency vehicle near the

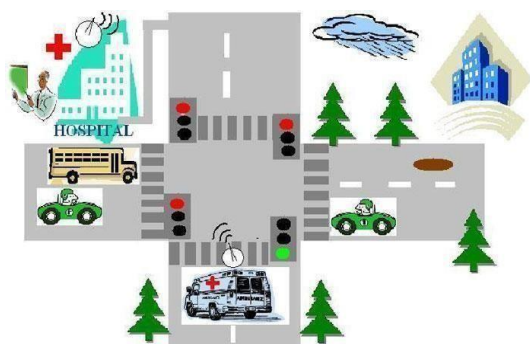


Fig : Proposed System

system and density of the vehicles is normal the signals function in a round robin way i.e one signal after the other.

In general we use ultrasonic sensors for object detection. The ultrasonic sensors send waves in all direction when these waves hit a object they reflect back indicating that an object is detected. We use this to know if there is any vehicle Ultrasonic sensors are used to detect objects in their way. We place an Ultrasonic seenscr at some distance from the signal. If vehicle is detected then it indicates that the lane has more density. As there is more density in that lane we provide extra time ensuring that the traffic does'nt get congested.

RFID technology comprises an RFID reader and an RFID card. The RFID card transmits a unique signal that is received by the reader. This technology can be used to aid emergency vehicles. RFID tags can be inserted into emergency vehicles and readers can be placed at certain distances along the route. When an emergency vehicle approaches the reader, the RFID tag triggers the green signal, allowing the vehicle to proceed without any obstacles. In summary, RFID technology can be used to facilitate the smooth movement of emergency vehicles.

IV. PROPOSED APPROACH: The main focus of the proposed approach is to efficiently manage traffic and reduce vehicle congestion to facilitate the movement of emergency vehicles and reduce their response time. Although some researchers have attempted to address traffic management through prioritizing cars, their solutions have not been adequate. Therefore, a new solution that employs IoT technology to help emergency vehicles navigate through traffic has been

proposed. This solution utilizes devices such as the RC522 RFID Module, RFID tag, Ohms Resistor, Signal Light Module, and Connection Wires to manage traffic signals. The proposed model aims to communicate with sensors in the vehicle and signal, enabling the traffic signal to change once an emergency vehicle arrives.

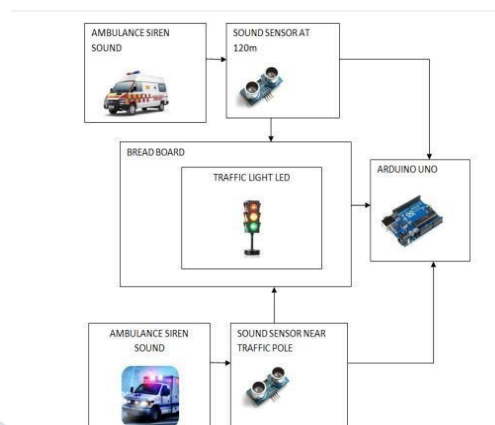


Fig : Framework for the system

RFID Tags: The use of smart barcodes is commonly seen in RFID tags, which function as tracking devices for identifying objects. These RFID tags operate using radio frequency technology, hence the name "radio frequency identification."

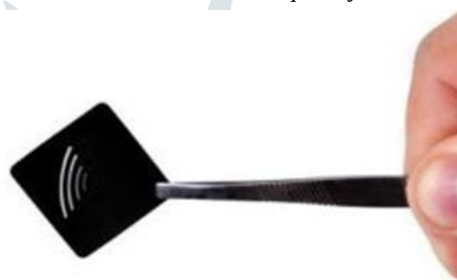


Fig RFID Tag

RFID Module: The RC522 is a highly integrated RFID card reader that operates on non-contact 13.56 MHz communication. It was created by NXP as a read-write chip with low power consumption, low cost, and a small footprint, making it the ideal option for smart meters and portable handheld devices.



Fig : RFID MODULE

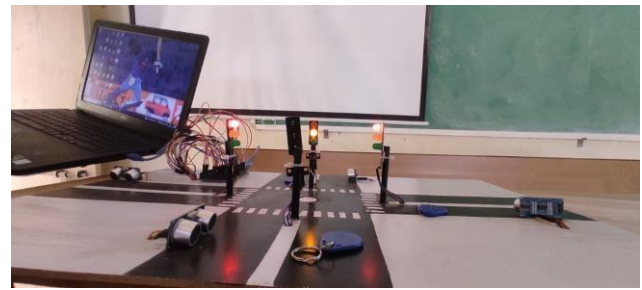


Fig Smart Traffic System

HC-SR04 Ultrasonic Sensor: The HC-SR04 is an inexpensive ultrasonic distance sensor that can accurately measure distances from 2 cm to 400 cm with an accuracy of up to 3mm. Each module comes equipped with an ultrasonic transmitter, receiver, and control circuit, enabling it to perform non-contact measurements.



Fig : Ultrasonic Sensor

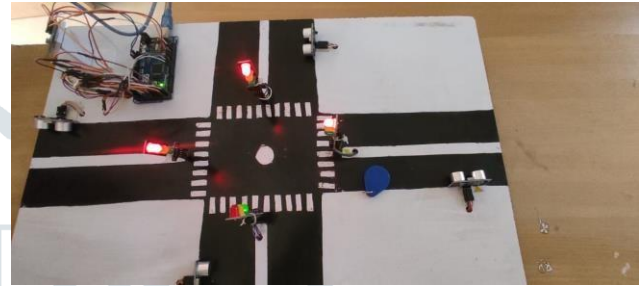


Fig – Output for high density vehicles

V. RESULTS

We have built a physical system with 4 lanes 4 signals and some toy vehicles to illustrate how the system works. In this chapter, we will see how our Smart Traffic System works.

A. Working:

1) Normal scenario:

In the normal scenario where our system doesn't have high traffic the system follows the old traffic system rules which is the round robin methodology.

2) High density scenario:

If there is high density at any lane the ultrasonic sensors come into work and provide some extra time for the lane so that the vehicles don't get into traffic congestion.

3) Emergency vehicle scenario:

If there is any emergency vehicle approaching the lane then the RFID reader sends an output signal to the green light in the signal.

B. Result Screenshot:

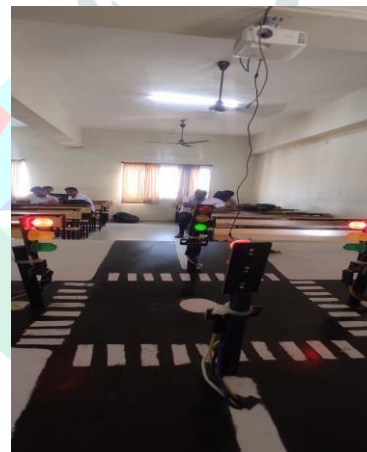


Fig – Output for emergency vehicles

In the above two images, we can see that we have two portals- class and exam.

VI. CONCLUSION:

From this project, we ensure effective working of Traffic System without having any congestion. By using IOT technology which includes Ultrasonic sensors and RFID technology we met the key requirements of this system which are:

- Density based traffic monitoring.
- Emergency vehicle detection.

By maintaining these constraints, we as a team are underway of building a Smart Traffic System which will ensure that the traffic doesn't get congested

A smart traffic management system using IoT can significantly enhance the clearance of emergency vehicles on the road, which is crucial in saving lives and reducing response time during emergencies. The system integrates different technologies, including GPS, sensors, and communication networks, to create an intelligent network that can provide real-time information about traffic flow, congestion, and accidents.

By using this system, emergency vehicles can be given priority access to the road and clear the path in front of them, avoiding delays caused by traffic congestion. This system can also communicate with other vehicles and devices, such as traffic lights and road signs, to ensure smooth and safe passage for emergency vehicles.

Overall, the implementation of a smart traffic management system using IoT can significantly improve the efficiency of emergency response services, thereby contributing to public safety and saving lives.

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